

KARUIZAWA, Japan

WMO: 476220

Lat: 36.3419N

Lon: 138.5472E

Elev: 1001

StdP: 89.87

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-10.0	-15.6	1.1	-4.4	-14.1	1.2	-4.3	6.1	-0.4	5.4	-0.5	1.6	0	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	28.6	21.8	27.3	21.4	25.8	20.8	22.9	26.5	22.2	25.6	21.6	24.6	2.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	18.6	24.6	21.2	17.9	23.9	20.6	17.3	23.1	73.5	26.7	70.7	25.7	68.2	24.8	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.2	4.5	4.1	DB	-15.0	30.8	1.7	1.0	-16.2	31.4	-17.2	32.0	-18.1	32.6	-19.3	33.3	
			WB	-15.3	24.3	1.7	0.5	-16.5	24.6	-17.5	24.9	-18.4	25.1	-19.7	25.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-3.2	-2.4	1.5	7.2	12.6	16.4	20.8	21.5	17.2	11.2	5.3	-0.1
	DBStd	9.07	2.84	3.38	3.72	3.89	2.94	2.76	2.57	2.30	3.25	3.11	3.18	3.11
	HDD10.0	1612	409	346	265	99	10	0	0	0	0	23	146	314
	HDD18.3	3610	667	579	523	333	177	71	9	5	58	223	392	573
	CDD10.0	1272	0	0	1	16	92	192	334	358	217	58	4	0
	CDD18.3	228	0	0	0	0	1	12	85	104	26	0	0	0
	CDH23.3	1014	0	0	0	3	22	52	398	466	74	1	0	0
	CDH26.7	172	0	0	0	0	1	4	76	86	6	0	0	0
Wind	WSAvg	1.8	2.1	2.1	2.2	2.1	1.9	1.5	1.5	1.5	1.5	1.6	1.8	2.0
Precipitation	PrecAvg	1206	38	40	64	79	102	145	172	151	195	126	60	33
	PrecMax	1490	95	85	119	175	220	237	371	303	417	373	171	93
	PrecMin	768	7	14	23	26	37	68	75	57	36	6	5	4
	PrecStd	185	25	22	25	39	44	45	74	69	86	82	39	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.9	12.4	17.6	22.8	25.8	27.1	30.2	30.4	27.5	22.4	17.6	12.1
		MCWB	4.6	6.5	8.0	12.4	15.2	19.3	21.7	21.8	21.3	16.8	11.1	7.3
	2%	DB	6.1	8.6	13.6	19.4	23.2	24.7	28.7	28.8	25.4	19.8	15.0	9.3
		MCWB	2.2	3.9	6.3	10.5	13.8	18.2	21.9	22.0	20.4	15.6	10.0	5.5
	5%	DB	4.0	5.7	11.0	17.2	21.2	23.2	27.3	27.5	23.7	18.0	12.8	7.5
		MCWB	0.4	1.7	5.2	9.4	13.0	17.5	21.6	21.8	19.5	14.2	8.8	3.9
	10%	DB	2.0	3.5	8.6	14.9	19.4	21.6	25.7	26.1	22.0	16.3	10.9	5.4
		MCWB	-1.0	0.3	4.1	8.8	12.5	16.9	21.1	21.5	18.9	13.4	7.9	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	7.8	9.7	15.0	17.9	21.2	23.9	23.9	22.1	18.3	13.1	9.2
		MCDB	7.7	11.3	14.1	17.6	21.1	24.7	27.9	27.8	25.6	20.7	15.5	10.2
	2%	WB	2.6	4.5	7.9	12.7	16.2	19.9	22.8	23.1	21.1	16.7	11.2	6.3
		MCDB	5.2	7.5	11.3	16.9	20.2	22.9	26.6	26.7	24.0	18.8	13.1	8.3
	5%	WB	0.7	2.3	6.2	11.0	14.9	18.9	22.1	22.4	20.4	15.3	9.9	4.4
		MCDB	3.4	4.8	9.9	14.6	18.6	21.5	25.6	25.8	22.7	16.9	11.5	6.5
	10%	WB	-0.6	0.6	4.5	9.6	13.8	17.9	21.4	21.8	19.6	14.0	8.6	2.7
		MCDB	1.6	3.1	7.9	13.1	17.3	20.5	24.5	24.9	21.4	15.4	10.5	5.1
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.5	11.4	12.5	11.7	8.8	8.0	8.4	7.8	9.1	10.2	9.6
		MCDBR	12.0	13.4	15.2	16.0	15.5	11.7	10.8	10.7	9.6	10.9	12.7	12.0
		MCWBR	8.3	9.0	8.9	8.3	7.6	5.8	5.1	5.0	5.0	6.5	7.9	8.1
	5% WB	MCDBR	10.6	11.9	13.3	12.5	12.5	9.1	9.6	9.4	8.4	8.8	10.1	10.7
		MCWBR	7.8	8.5	8.9	7.7	7.0	5.0	4.9	4.8	4.6	5.7	7.1	7.9
Clear-Sky Solar Irradiance	taub	0.314	0.348	0.402	0.444	0.469	0.504	0.500	0.504	0.438	0.390	0.341	0.315	
	taud	2.238	2.139	2.078	2.056	2.038	2.007	2.095	2.074	2.228	2.292	2.385	2.352	
	Ebn at Noon	866	875	857	840	823	792	792	779	815	823	835	845	
	Edh at Noon	104	130	152	165	170	176	160	160	130	111	90	88	
All-Sky Solar Radiation	RadAvg	2.38	3.14	4.03	4.80	5.28	4.77	4.70	4.70	3.80	3.15	2.51	2.13	
	RadStd	0.12	0.20	0.26	0.39	0.52	0.32	0.66	0.44	0.33	0.31	0.19	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page